BERT and Language Models

What Is a Language Model?

A language model assigns probabilities to sequences of tokens. Traditional n-gram language models estimate these probabilities from token co-occurrence counts. Neural language models use neural networks to learn dense representations that generalize far better to unseen sequences.

Modern large language models (LLMs) are trained on massive text corpora and learn to represent language at a deep semantic and syntactic level, enabling transfer to a wide range of downstream tasks with little task-specific training.

Pre-Training and Fine-Tuning

The pre-train then fine-tune paradigm is central to modern NLP. A model is first pre-trained on a large general corpus using a self-supervised objective, learning rich language representations. It is then fine-tuned on a smaller labeled dataset for a specific task.

This approach is powerful because the pre-trained representations encode general linguistic knowledge that transfers across tasks, reducing the need for large labeled datasets for every new task.

BERT: Bidirectional Encoder Representations from Transformers

BERT (Devlin et al., 2018) was a landmark model that demonstrated the power of deeply bidirectional pre-training. Unlike previous models that processed text left-to-right or combined shallow left-to-right and right-to-left models, BERT conditions on both left and right context simultaneously in every layer.

BERT uses the Transformer encoder architecture. The base model has 12 layers, 768 hidden dimensions, and 12 attention heads (110M parameters). The large model has 24 layers, 1024 dimensions, and 16 heads (340M parameters).

BERT Pre-Training Objectives

BERT is pre-trained using two objectives. The first is Masked Language Modeling (MLM): 15% of input tokens are randomly masked, and the model must predict the original tokens from context. This forces bidirectional reasoning since the model cannot simply copy from left to right.

The second objective is Next Sentence Prediction (NSP): given two sentences A and B, the model predicts whether B actually follows A in the original text. This was intended to improve performance on tasks requiring sentence-pair reasoning, though later work found NSP to be less useful than originally thought.

Input Representation

BERT's input combines token embeddings, segment embeddings (indicating sentence A or B), and positional embeddings. A special [CLS] token is prepended to every input; its final hidden state is used as the aggregate representation for classification tasks.

A [SEP] token separates sentences and marks the end of input. The tokenizer uses WordPiece subword tokenization, splitting rare words into subword units to handle out-of-vocabulary words gracefully.

Fine-Tuning BERT

Fine-tuning BERT for downstream tasks is straightforward. For classification, a linear layer is added on top of the [CLS] token representation. For token-level tasks like named entity recognition, predictions are made from each token's representation. For question answering, the model predicts start and end token positions.

BERT achieved state-of-the-art results on 11 NLP benchmarks when released, including GLUE, SQuAD, and MultiNLI, demonstrating the broad transferability of its representations.

Variants and Successors

RoBERTa (Liu et al., 2019) showed that BERT was undertrained. By training longer, on more data, with larger batches, removing NSP, and using dynamic masking, RoBERTa significantly outperformed BERT.

DistilBERT uses knowledge distillation to produce a 40% smaller model that retains 97% of BERT's performance, making it practical for deployment.

Domain-specific variants like BioBERT (biomedical), SciBERT (scientific), and FinBERT (finance) demonstrate the value of continued pre-training on domain-specific corpora.

Encoder-only models like BERT remain widely used for retrieval, classification, and embedding tasks, even as decoder-only models like GPT dominate generation tasks.